

Java Notes (Enhanced with One-Liners + Diagrams)

 Source:

◆ 1. String in Java

One-Line Definition

👉 String is an immutable object that represents a sequence of characters.

◆ Diagram: String Memory (String Pool)

String Pool (Heap Area)

| "Java" | "Hello" |

↑ ↑
| |
s1, s2 s3

👉 Same value → Same memory reference

◆ Diagram: Immutable Nature

Before:

s = "Hello"

After:

s.concat(" World")

Memory:

"Hello" "Hello World"

↑ ↑
s (new object)

👉 s still points to "Hello"

◆ Quick Concept

- ✓ Stored in String Pool
 - ✓ Immutable
 - ✓ Use `.equals()` for comparison
-

◆ Advantages

- Secure & thread-safe
- Memory efficient
- Easy to use

◆ Disadvantages

- Slow for modification
 - Extra objects created
 - Memory overhead
-

◆ 2. Object Class

One-Line Definition

👉 Object class is the root superclass of all Java classes providing common methods.

◆ Diagram: Object Hierarchy

Object (Top Most Class)

↑

| | |
String Student Thread

👉 Every class inherits Object

◆ Diagram: equals() vs ==

s1 ----> "Java" (Heap)

s2 ----> "Java" (Heap)

s1 == s2 ❌ (Different memory)

s1.equals(s2) ✅ (Same content)

◆ Advantages ✅

- Common methods
- Supports polymorphism
- Code reuse

◆ Disadvantages ❌

- Needs overriding
 - Default behavior not always useful
-

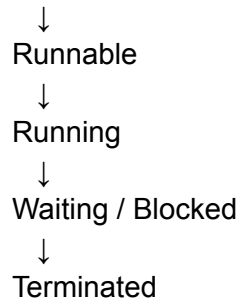
◆ 3. Threads in Java

✅ One-Line Definition

👉 Thread is a lightweight process that allows concurrent execution of tasks.

◆ Diagram: Thread Lifecycle

New



◆ Diagram: Multithreading Concept

Single Thread:

[Task1 → Task2 → Task3]

Multithreading:

[Task1]

[Task2] → Run in Parallel

[Task3]

◆ Advantages ✓

- Faster execution
- Better CPU utilization
- Responsive apps

◆ Disadvantages ✗

- Complex
- Deadlock
- Debugging difficult

◆ 4. Wrapper Classes

✓ One-Line Definition

👉 Wrapper classes convert primitive data types into objects.

◆ Diagram: Primitive vs Wrapper

Primitive		Wrapper
int	----->	Integer
char	----->	Character
double	----->	Double

◆ Diagram: Autoboxing & Unboxing

int a = 10;

Autoboxing:

Integer obj = a;

Unboxing:

int b = obj;

◆ Advantages

- Works with Collections
- Utility methods
- Supports null

◆ Disadvantages

- Slower than primitives
 - More memory usage
 - Boxing overhead
-

★ Ultra Short Revision (Exam Ready)

👉 String → Immutable, Pool

👉 Object → Parent class

👉 Thread → Parallel execution

👉 Wrapper → Primitive → Object